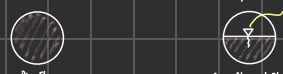
HYDRAULICS

Pipe Flow: The movement of a fluid through a closed conduit (eg. pipe, duct, tube)

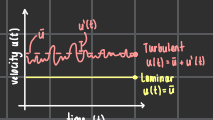
→ If the closed conduit is not full, it is called a "channel" (open) flow.



Pipe Flow
→ The driving force is pressure gradient (Δp)

Open Channel Flow
→ The driving force is gravity

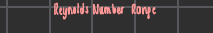
The flow of a fluid in a pipe can either be turbulent or laminar.



Reynolds Number: $Re = \frac{\rho U L}{\mu} = \frac{UL}{\nu}$ (fluid density, velocity, characteristic length (pipe diameter in pipe flow), dynamic viscosity, kinematic viscosity $\nu = \frac{\mu}{\rho}$)

$R_e = \frac{\rho U L}{\mu} = \frac{\text{inertial forces}}{\text{viscous effects}}$ ~ momentum = mass x velocity ~ frictional forces due to fluid viscosity

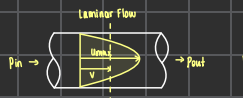
Reynolds Number Range



(For Pipe Flow)

* If the viscous forces are dominant → Laminar Flow
* If the inertial forces are dominant → Turbulent Flow

$\Delta p = p_{in} - p_{out}$ ⇒ laminar $\Delta p <$ turbulent Δp



→ Viscous effects decrease towards the axis → u_{max}
→ Velocity profile is parabolic
→ $V = 0.5\, u_{max}$

→ Velocity distribution is more uniform and flatter due to the momentum interchange between fluid layers
→ $V = 0.9\, u_{max}$ (fully developed f.f.)

Darcy-Weisbach Eq.: $\frac{\Delta P}{L} = f \frac{\rho}{2} \frac{V_{ave}^2}{D}$ (friction factor)

f → laminar f.: $f = 64/Re$
→ turbulent f.: $\frac{1}{\sqrt{f}} = -2\log\left(\frac{\varepsilon}{3.7D} + \frac{2.51}{Re\sqrt{f}}\right)$ (Colebrook Equation)

$\frac{\varepsilon}{D}$: relative roughness

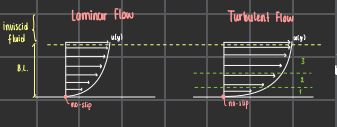
* we use the moody diagram

Momentum Transport

Laminar	Turbulent
→ Molecular Viscosity: Momentum transfer is caused by molecular interaction	→ Molecular Viscosity: Momentum transfer is caused by molecular interaction → Turbulent Mixing: Momentum transfer is caused by fluctuating fluid motion and turbulent eddies

Boundary Layer: A boundary layer develops when viscous effects near a boundary creat a velocity gradient.

↳ no-slip condition: $u = 0$ at the walls, farther away from the wall, $u \to u_\infty$



* There are 3 regions to the boundary layer in turbulent flows

1. Viscous (Laminar) Sublayer: Molecular Viscosity > Turbulent Mixing , Thickness ↓ Re ↑ , $\tau = \mu \frac{du}{dy}$

2. Buffer Layer: Molecular Viscosity ≅ Turbulent Mixing (↑) , $\tau = \mu \frac{du}{dy} + (-\rho \overline{u'v'})$

3. Logarithmic Region: Turbulent Mixing > Molecular Viscosity , $\tau = -\rho \overline{u'v'}$

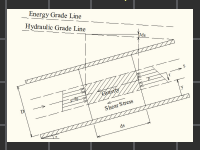
* Hydraulically Smooth Pipe: Thickness of Viscous Sublayer > Roughness Height
* Hydraulically Rough Pipe: Thickness of Viscous Sublayer < Roughness Height
* Transition Pipe Flow: Thickness of Viscous Sublayer ≈ Roughness Height

* If the roughness of the surface is entirely within the viscous sublayer(↑), it is said to be a hydraulically smooth surface.
* When reviewing the moody diagram, at higher R_e the thickness of the viscous sublayer is extremely thin, so the effect of the surface roughness is governing.

When a fluid enters a pipe, it has a nearly uniform velocity profile. As the fluid moves along in the pipe, viscous effects cause a laminar boundary layer to form. The boundary layer in a pipe can grow only as thick as the pipe radius. If this boundary layer reaches its max. thickness (boundary layers meet at the centre of the pipe) before transition occurs (from laminar to turbulent), the rest of the flow is said to be laminar flow, if not, it is said to be a turbulent flow.

* The transition from laminar to turbulent flow is mainly caused by the following conditions: surface roughness, vibrations, inlet conditions etc.

Distribution of Velocity in Pipe Flow: There are 3 forces acting on a cylindrical fluid element, 1- Pressure, 2- Gravity, 3- Shear along perimeter



* We write an eq. for a steady flow (no accelerat.)

$\Sigma F = ma$

$(p+dp)\pi r^2 - p\pi r^2 - (\rho g \pi r^2 ds)\sin\alpha - \tau_y (2\pi r ds) = 0$ ⇒ $\tau_y = \left(\frac{1}{\rho}\frac{dp}{ds} - \sin\alpha\right)\frac{\rho g r}{2}$

pressure force | gravity force | shear force

* energy loss (head loss) due to friction,

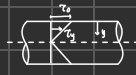
$\Delta h_L = \left(\frac{V^2}{2g} + \frac{(p+dp)}{\delta} + z_1\right) - \left(\frac{V^2}{2g} + \frac{p}{\delta} + z_2\right) + \frac{dp}{\delta} + \frac{(z_1 - z_2)}{ds}\sin\alpha$

$\Rightarrow \frac{\Delta h_L}{ds} = J = \frac{1}{\rho g}\frac{dp}{ds} - \sin\alpha$ ↝ energy loss per unit length

combine both equations ⇒ $\tau_y = \rho g\left(\frac{p}{\gamma} - \frac{J}{2}\right)r$, $r = \frac{D}{2} - y$

* Hydraulic Radius: $R = \frac{\text{Flow Area}}{\text{Wetted Perimeter}}$

→ for a circular cross-section: $R = \frac{\pi D^2/4}{\pi D} = D/4$

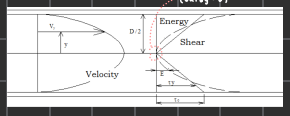


$\tau_0 = \rho g R J$

* $\tau_{Total} = \tau_{laminar} + \tau_{turbulent}$, $\tau_l = \rho\nu\frac{du}{dy} = \mu\frac{du}{dy}$, $\tau_t = -\rho\overline{u'v'}$ (fluctuation velocity)

Laminar Flow Through Pipes

Kinematic viscosity: $\nu = \frac{\tau_y}{\rho}\frac{du}{dy}$, $\tau_y = \mu\frac{du}{dy}$ (Newtonian fluids and laminar flow)



$u = \frac{\rho J}{\nu}\left(\frac{D}{4} - \frac{y^2}{D}\right) \Rightarrow u_{axis} = u_{max} = \frac{\rho J}{\nu}\frac{D^2}{16} = \frac{J}{\nu}R^2$ ($y = D/2$)

$V = 0.5\, u_{max} = \frac{\rho J}{\nu}\frac{D^2}{32}$ (circular cross-section)

$= \frac{J}{\nu}\frac{R^2}{2}$ (arbitrary cross-section)

Average

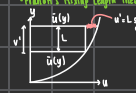
} Hagen-Poiseuille Equation

$Q = VA = \frac{\pi D^4 J}{128\mu}$, $J = \frac{\Delta P/\gamma}{L} \Rightarrow \Delta p = \frac{128\mu L Q}{\pi D^4}$

Turbulent Flow Through Pipes

Momentum is transfered by the mixing of adjacent layers (eddy movements). Hence the shear stress will be proportional to the square of the velocity gradient.

- Prandtl's Mixing Length Theory: Mixing length(L) is the distance that a fluid particle moves without loosing its characteristics in a turbulent flow.



$u' = L\frac{d\bar{u}}{dy}$ ⇒ $\bar{u}(y+L) = \bar{u}(y) + L\frac{d\bar{u}}{dy} + \frac{L^2}{2!}\frac{d^2\bar{u}}{dy^2}$ (Taylor's Exp.)

* Strong mixing of the fluid creates a more uniform velocity gradient.

* Fluctuating velocity component in the main flow direction, and that of perpendicular direction are assumed equal. ($u' = v'$) ⇒ $u' = L\frac{d\bar{u}}{dy} = v'$ ⇒ $\overline{u'v'} = -L^2\left(\frac{d\bar{u}}{dy}\right)^2$

* A fluid particle,
→ fluctuating in (+)y direction decelerates flow velocity in the x direction. → $v'(+)u'(-)$ ⇒ $\overline{u'v'} < 0$, shear stress has to be (+) ⇒ $\tau_t = -\rho\overline{u'v'}$
→ " " (−) " " accelerates " " " " " . → $v'(-)u'(+)$

$\tau_T = \tau_l + \tau_t$

$\rho\nu\frac{du}{dy}$, $\rho L^2\left|\frac{d\bar{u}}{dy}\right|\frac{d\bar{u}}{dy}$

- Von Karman: $L = \kappa y$, if $\frac{du}{dy} > 0 \rightarrow \tau = \rho\kappa^2 y^2\left(\frac{d\bar{u}}{dy}\right)^2$, $L^2 = (\tau_y/\rho)/(du/dy)^2 \Rightarrow \sqrt{\tau_y/\rho} = L(du/dy)$

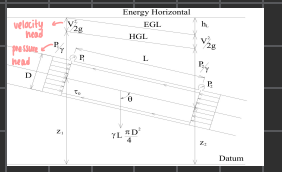
V.K. coefficient (kappa)

↳ if L increases, the velocity gradient decreases
↳ increase of L causes strong mix

$\varepsilon = L^2\left(\frac{du}{dy}\right)$ eddy (turbulent) viscosity , $\tau_t = \rho\varepsilon\left(\frac{du}{dy}\right)$

Energy Loses in Pipes

a) Head Loss Due to Friction
b) Local Loss (Minor Loss)



a) Head Loss Due to Friction: Mechanical Energy —friction→ Heat Energy

energy loss: $\Delta h_L = \left(\frac{P_1}{\gamma} + \frac{V_1^2}{2g} + z_1\right) - \left(\frac{P_2}{\delta} + \frac{V_2^2}{2g} + z_2\right)$ * $V_1 = V_2$ (constant diameter)

$\Delta h_L = \left(\frac{P_1}{\gamma} + z_1\right) - \left(\frac{P_2}{\gamma} + z_2\right)$

* steady flow ⇒ acceleration is zero , $\Sigma F = ma = 0$,

$P_1\frac{\pi D^2}{4} - P_2\frac{\pi D^2}{4} + \gamma L\frac{\pi D^2}{4}\cos 0 - \tau_0 \pi D L = 0 \Rightarrow \tau_0 = \frac{D\gamma}{4L}\left(\frac{P_1}{\delta} + z_1 - \frac{P_2}{\delta} - z_2\right)$

$\tau_0 = \frac{D\gamma}{4L}\Delta h_L$ we can obtain the non-dimensional frictional factor λ by dividing this by $\rho V^2/8$,

$\Delta h_L = \lambda\frac{L}{D}\frac{V^2}{2g}$ Darcy-Weisbach Equation

* The Darcy-Weisbach equation applies for all. The friction coefficient λ, in general, an equation of both the Reynolds number and the relative roughness.

Friction Coefficient (λ)	
Laminar	$\lambda = f(Re)$
Turbulent	
Hydraulic Smooth Pipe	$\lambda = f(Re)$
Hydraulic Rough Pipe	$\lambda = f(L/D)$
In Transition	$\lambda = f(Re, L/D)$

i) For Laminar Flow

Average Velocity, $V=(\rho T/\nu)(D^2/32)$

$(T=\Delta h_L/L)$ $V=\frac{\rho}{\nu L}\cdot\frac{D^2}{32}\rightarrow \Delta h_L=\frac{64\nu}{VD}\cdot\frac{L}{D}\cdot\frac{V^2}{2g}\Rightarrow \boxed{\Delta h_L=\frac{64}{Re}\frac{L}{D}\frac{V^2}{2g}}$, $\boxed{\lambda=\frac{64}{Re}}$

ii) For Turbulent Flow

a- Hydraulic Smooth Pipe: $\frac{V}{u_*}=5.75\log\left(\frac{Du_*}{2\nu}\right)+1.73$, $\lambda=8Tg/\rho V^2$

$$\boxed{\sqrt{\frac{8}{\lambda}}=5.75\log\left(\frac{VD}{\nu}\sqrt{\frac{\lambda}{32}}\right)+1.73}$$

b- Hydraulic Rough Pipe: $\frac{V}{u_*}=5.75\log\left(\frac{8}{2k}\right)+4.75$, $\frac{V}{u_*}=\sqrt{\frac{8}{\lambda}}$

$$\boxed{\sqrt{\frac{8}{\lambda}}=5.75\log\frac{D}{2k}+4.75}$$

c- Transition Regime (semi-empirical formula) Colebrook - White Equation

$$\boxed{\frac{1}{\sqrt{\lambda}}=-2\log\left(\frac{k}{3.7D}+\frac{2.51}{Re\sqrt{\lambda}}\right)}$$

Minor Head Losses $K\frac{V_1^2}{2g}$

1. Sudden Expansion: $\Delta h_{SE}=K_{SE}\frac{V_1^2}{2g}=\frac{(V_1-V_2)^2}{2g}$

2. Entrance Into Reservoir: $\Delta h_E=K_E\frac{V_1^2}{2g}$

3. Sudden Contraction: $\Delta h_{SC}=K_{SC}\frac{V_2^2}{2g}$

4. Entrance to a Pipeline from a Reservoir: $\Delta h_C=K_C\frac{V_1^2}{2g}$

5. Pipe Fittings: $\Delta h_{PF}=K_{PF}\frac{V^2}{2g}$

6. Bends: $\Delta h_B=K_B\frac{V^2}{2g}$

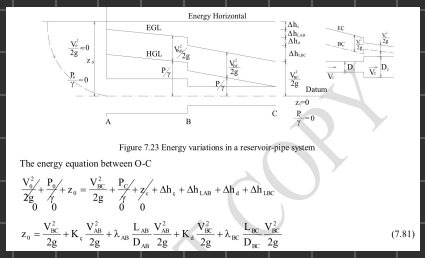


Figure 7.23 Energy variations in a reservoir-pipe system

The energy equation between O-C

$$\frac{V_O^2}{2g}+\frac{P_O}{\gamma}+z_O=\frac{V_{DC}^2}{2g}+\frac{P_C}{\gamma}+z_C+\Delta h_e+\Delta h_{LAB}+\Delta h_d+\Delta h_{LDC}$$

$$z_O=\frac{V_{DC}^2}{2g}+K_e\frac{V_{AB}^2}{2g}+\lambda_{AB}\frac{L_{AB}}{D_{AB}}\frac{V_{AB}^2}{2g}+K_d\frac{V_{DC}^2}{2g}+\lambda_{DC}\frac{L_{DC}}{D_{DC}}\frac{V_{DC}^2}{2g} \qquad (7.81)$$

Various Pipe Problems

1. Uniform Flow in a Constant Diameter Pipe Exiting from a Reservoir

Single Straight Pipeline: $z_O=\frac{V_1^2}{2g}+\frac{P_1}{\gamma}+\Delta h_L+\Delta h_C$

2. Flow Between Reservoirs: $z_O=\Delta h_C+\Delta h_L+\Delta h_E=\Sigma\Delta h$

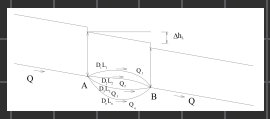
3. Long Pipes: $L\gg D,V^2/2g$ could be ignored $\Rightarrow$ EGL and HGL coincide. $z_O=\Delta h_L$ (only friction loss-major loss)

4. Pipe in Series: The head of one is connected to the tail of another one.
- The head loss is cumulative: $z_O=\Delta h_C+\Delta h_{L_{AB}}+\dots=\Sigma\Delta h$
- Discharge through each pipe is equal: $\Theta=V_{AB}A_{AB}=V_{BC}A_{BC}=V_{CD}A_{CD}$

1- Given: Flow Rate + Pipe Combinations ; Find: Total Head Loss

2- Given: Allowable Head Loss + Pipe Combinations ; Find: Flow Rate

5. Pipes in Parallel: Pipes branch out in diverging nodes and join at converging nodes.
- The head loss through each pipe is the same = difference in total energy between diverging node and converging node.
- Discharge this cumulative $\Rightarrow$ Distributed among the branching pipes.

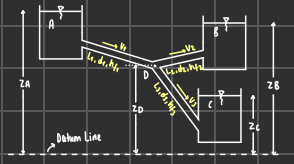


a) total energy between A-B: constant $\Rightarrow$ head loss through each pipe is same. $\Delta h_{L1}=\Delta h_{L2}=\dots=\Delta h_{Ln}$

b) Continuity equation: $\Theta=\Theta_1+\Theta_2+\dots+\Theta_n$

c) energy eq. (local h.L's are ignored): $\frac{P_A}{\gamma}+\frac{V_A^2}{2g}+z_A=\frac{P_B}{\gamma}+\frac{V_B^2}{2g}+z_B+\Delta h_L$ (for long p.s., $V^2/2g$ is neglected)

6. Branching Pipes: At least 3 reservoirs are connected with each other through a piping system with at least 1 junction. (assume to neglect minor losses)



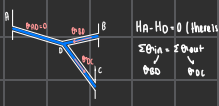
$\rightarrow$ total energy at the surface of each reservoir is taken as the elevation of the fluid level.

1. Σ Liquid entering junction = Σ liquid leaving junction $\Rightarrow \Sigma\Theta=0$

2. All pipes meeting at junction D have the same pressure at D.

* If the pressure head (z_D+P_D/γ) is known at junction D, the flow direction can be obtained. If not, following steps should be done.

Assumption 1 - assume height of the total energy line at D (generally equal to that of A), $z_A=z_D+\frac{P_D}{\gamma}\Rightarrow$ if $\Theta_{BD}=\Theta_{DC}$ assumption is correct; if not, (A-1, A-2)



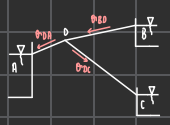
$H_A-H_D=0$ (there is no flow in between A and D)

$\Sigma\Theta_{in}=\Sigma\Theta_{out}$

Θ_{BD} Θ_{DC}

If $\Theta_{BD}\neq\Theta_{DC}$,

Assumption 2 - $\Theta_{BD}>\Theta_{DC}$: the flow is from B to A and C $\Rightarrow$ head of junction D > the elevation of reservoir A.



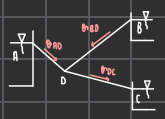
$\Theta_{BD}=\Theta_{DA}+\Theta_{DC}$

$z_B=z_A+\Delta h_{LBD}+\Delta h_{LDA}$

$=z_C+\Delta h_{LBD}+\Delta h_{LDC}$

$*z_D=\frac{P_D}{\gamma}=z_A+\Delta h_{LDA}$

Assumption 3 - $\Theta_{BD}<\Theta_{DC}$: the flow is from B and A to C $\Rightarrow$ total head of junction D < the elevation of reservoir A.

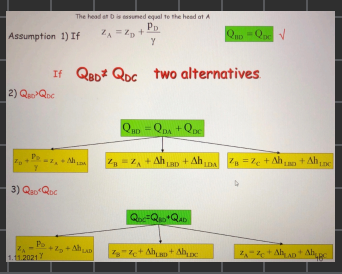


$\Theta_{DC}=\Theta_{BD}+\Theta_{AD}$

$z_B=z_C+\Delta h_{LBD}+\Delta h_{LDC}$

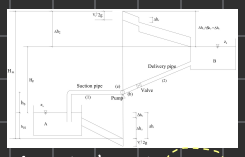
$=z_C+\Delta h_{LAD}+\Delta h_{LDC}$

$=z_A-\frac{P_D}{\gamma}=z_D+\Delta h_{LAD}$



COPY

7. Simple Pump in a Pumpline: The source of water is lower than the area we need to distribute to, so the head is developed by a pump.



V_s: velocity in suction pipeline (As)

V_d: velocity in delivery pipeline (Ad)

Δh_s: total head loss in suction pipeline

Δh_d: head loss in delivery pipe (friction, valves, etc.)

ΔH_g: static lift (elevation head)

ΔH_m: manometric head → difference between the pumps inlet (a) and outlet (b) points

$$H_m = \underbrace{\frac{P_b}{\delta} + \frac{V_b^2}{2g}}_{\text{right hand-side of pump}} - \underbrace{\left(\frac{P_a}{\delta} + \frac{V_a^2}{2g}\right)}_{\text{left hand-side of pump}}$$

A-a: $z_A = \frac{P_a}{\delta} + \frac{V_a^2}{2g} + \Delta h_s \Rightarrow \frac{P_a}{\delta} = z_A - \frac{V_a^2}{2g} - \Delta h_s$

B-b: $\frac{P_b}{\delta} = z_B + \Delta h_d - \frac{V_b^2}{2g}$

$$H_m = z_B + \Delta h_d - \frac{V_b^2}{2g} - \left(z_A - \frac{V_a^2}{2g} - \Delta h_s + \frac{V_a^2}{2g}\right)$$

$$H_m = z_B - z_A + \Delta h_d + \Delta h_s$$

$$H_m = H_g + \Delta h_d + \Delta h_s$$

~ the power of pump which gives discharge, Θ, N: $N = \frac{\gamma H_m \Theta}{\eta}$ (kW)

γ: kN/m³, H_m: m, Θ: m³/s, η: efficiency

Hardy Cross Method: An iterative method for determining the flow in pipe network systems where the inputs and outputs are known, but the flow inside the network is unknown.

* Network of Pipes: Interconnected pipes through which the flow of a given outlet may come from several circuits.

* Node: A point at which more than 1 pipe meets.

* Loop: A closed conduit consisting of more than 1 pipe.

~Following must be satisfied in a network of pipes,

1- Sum of the discharges entering and leaving a node is zero. $\Sigma\Theta_{node} = 0$

2- Sum of the total head loss around each circuit must be zero. $\Sigma_{loop}\, h_f = 0$

3- The Darcy-Weisbach eq. must be satisfied for each pipe.

Step

1- Assume the best distribution of flows that satisfies continuity at each node.

2- For each pipe, head losses are calculated $\Delta h_L = r.\Theta^n$ (for Darcy-Weisbach: n=2, for Williams-Hazen: n=1,85)

3- For each loop, sum the net head loss ($\Sigma \Delta h_L = \Sigma r\Theta^n$)

→ if $\Sigma \Delta h_L \approx 0$, assumed discharges are correct

→ if $\Sigma \Delta h_L \neq 0$, assumed discharges are not correct ⇒ for each circuit the discharges should be corrected by $\Delta\Theta$

4- $\Delta\Theta$: correction discharge, Θ_0: assumed initial discharge, Θ: corrected discharge

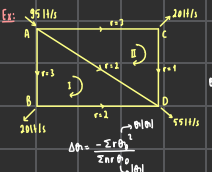
for each pipeline, the head loss is, $\Delta h_L = r.(\Theta_0 + \Delta\Theta)^n$ we are going to ignore the higher order terms

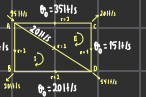
$\Sigma \Delta h_L = \Sigma r\Theta^n = \Sigma r\Theta_0^n + \Delta\Theta \Sigma n r\Theta_0^{n-1} = 0 \Rightarrow$

$$\Delta\Theta = \frac{-\Sigma r\Theta_0^n}{\Sigma n r\Theta_0^{n-1}} = \frac{-\Sigma \Delta h_L}{n \Sigma \Delta h_L / \Theta_0}$$

* $\Delta\Theta$ is added to flows in clockwise direction, subtracted from counterclockwise direction flows.

Ex:





$\Theta_0 = 35$ lt/s

$\Theta_0 = 40$ lt/s

$\Theta_0 = 15$ lt/s

$\Theta_0 = 20$ lt/s

1. döviler hakkında kısaca yorumlar yapınız

$$\Delta\Theta = \frac{-\Sigma r\Theta_0^2}{\Sigma n r\Theta_0}$$ lt/s

1. Loop: $\Sigma r\Theta^2$ | $\Sigma n r\Theta_0$ (0 alınış; n=2)

loop'un tersi yönünde			
AB	$-3\times40^2 = -4800$	$2\times3\times40 = 240$
BD	$-2\times20^2 = -800$	$2\times2\times20 = 80$
DA	$2\times10^2 = 200$	$2\times2\times10 = 40$
	-4800	400

$$\Delta\Theta_1 = -\frac{-4200}{400} = 12 \text{ lt/s}$$

Problem 7.1: A horizontal pipeline has a diameter of 0.1 m. Water is pumped through this pipeline with a mean velocity V=1.35 m/sec. Kinematic viscosity index of water $\nu=10^{-7}$ m^2/sec. Roughness of the boundary a=1.5 10^{-4} m.

a) What difference in head can be read from two manometers that are connected to the pipe and of which the distance between the points of connection equals to 100 m.

b) Same question for a liquid with a viscosity index. $\nu=10^{-5}$ m^2/s

c) Same question for a liquid with a viscosity index. $\nu=10^{-4}$ m^2/s

1. Akım türbülant mı, laminar mı? ($Re = VD/\nu$)
   - $Re < 2320$: Laminar ($V = 0.5 u_{max}$)
   - $Re > 2320$: Turbulent ($V = 0.81 u_{max}$)
2. for turbulent, thickness, $Re_* = \frac{u_* \delta_a}{\nu}$ ⇒ $\delta_a = 11.6\,\nu / u_*$

Viscous Sublayer	Overlap Layer	Outer Layer
$\tau = \mu \frac{du}{dy}$	$\tau = \mu \frac{du}{dy} - \rho \overline{u'v'}$	$\tau = -\rho \overline{u'v'} = \rho \varepsilon \frac{du}{dy}$

3.

Hydraulically Smooth	Transition	Hydraulically Rough
$k < \delta_a$	$\delta_a < k < 6\delta_a$	$k > 6\delta_a$
→ C depends on: ν	→ C depends on: ν + k	→ C depends on: k
→ $\frac{V}{u_*} = 1.75 + 5.75 \log \frac{u_* D}{2\nu}$	→ $\frac{V}{u_*} = 5.75 \log \frac{12R}{k + 0.35\delta_a}$ (hepsi için geçerli)	→ $\frac{V}{u_*} = 4.75 + 5.75 \log \frac{D}{2k}$

a) $\nu = 10^{-6}\ m^2/s$, $Re = \frac{VD}{\nu} = \frac{1.35 \times 0.1}{10^{-6}} = 1.35 \times 10^5 > 2320$: Turbulent ⇒ Chezy kullanılır

! Hydraulically rough kabulü, Δh bulmak için Δh'ı bilmemiz gerektiğinden sorunun kalanını bu varsayımla çözüp en son teyit edeceğiz. ($R = D/4 = 0.025m$)

$k > 6\delta_a$, $V = C\sqrt{RJ}$ — Chezy

$u_* = \sqrt{gRJ} = 3.13\sqrt{RJ}$ $\quad k = 2a$, $\frac{6R}{a + \delta_a/7}$

$\frac{V}{u_*} = 5.75 \log\left(\frac{12R}{k + 0.35\delta_a}\right) \Rightarrow V = (3.13\sqrt{RJ}) \times 5.75 \log\left(\frac{12R}{k + 0.35\delta_a}\right) = C\sqrt{RJ}$

$\Rightarrow C = 18 \log \frac{6R}{a} = 18 \log \frac{6 \times 0.1/4}{1.5 \times 10^{-4}} = 54\ m^{1/2}/s$, $V = C\sqrt{RJ}$, $\Delta h = JL$; $\Delta h = \frac{LV^2}{C^2 R} = \frac{100 \times 1.35^2}{54^2 \times 0.025} = 2.5m$

! başta yaptığımız kabulü kontrol et, $k > 6\delta_a$

$J = \Delta h / L = 2.5/100 = 0.025$

$u_* = \sqrt{gRJ} = \sqrt{10 \times 0.025 \times 0.025} = 0.079\ m/s$

$\delta_a = 11.6\,\nu / u_* = 11.6 \frac{10^{-6}}{0.079} = 1.47 \times 10^{-5} m$

$k = 3 \times 10^{-4}$, $6\delta_a = 8.8 \times 10^{-5}$; $k > 6\delta_a$ ✓ kabul doğru

b) $\nu = 10^{-5}\ m^2/s$: $Re = VD/\nu = \frac{1.35 \times 0.1}{10^{-5}} = 1.35 \times 10^4 > 2320$: Turbulent

ezâce bulduğumuz değeri kayıp deniyoruz ; $\Delta h = 2.5m$, $u_* = 0.079\ m/s$

$\delta_a = 11.6 \frac{10^{-5}}{0.079} = 1.47 \times 10^{-3} \rightarrow 6\delta_a = 8.8 \times 10^{-3} m > k = 3 \times 10^{-4}$ ⇒ ya transition'da ya da hydraulically smooth

! burada, verdiğimiz Δh değerine kabul görecek kadar yakınlaşana dek iterasyon yapılır.

1. iterasyon : $\Delta h = 3m$ (uymayan Δh değerinden biraz fazla bir değer denenir)

$J = \frac{\Delta h}{L} = \frac{3}{100} = 0.03$

$u_* = \sqrt{gRJ} = \sqrt{10 \times 0.025 \times 0.03} = 0.087\ m/s$

$\delta_a = 11.6\,\nu / u_* = 11.6 \times 10^{-5} / 0.087 = 1.34 \times 10^{-3} m$

$C = 18 \log \frac{6R}{a + \delta_a/7} = 18 \log \frac{6 \times 0.025}{1.5 \times 10^{-4} + 1.3 \times 10^{-3}/7} = 47.6$

$\Delta h = \frac{LV^2}{C^2 R} = \frac{100 \times 1.35^2}{47.6^2 \times 0.025} = 3.22m$

2. iterasyon : $\Delta h = 3.22m$

$J = \frac{\Delta h}{L} = \frac{3.22}{100} = 0.0322$

$u_* = \sqrt{gRJ} = \sqrt{10 \times 0.025 \times 0.0322} = 0.09\ m/s$

$\delta_a = 11.6\,\nu / u_* = 11.6 \times 10^{-5} / 0.09 = 1.29 \times 10^{-3} m$

$C = 18 \log \frac{6R}{a + \delta_a/7} = 18 \log \frac{6 \times 0.025}{1.5 \times 10^{-4} + 1.23 \times 10^{-3}/7} = 47.7$

$\Delta h = \frac{LV^2}{C^2 R} = \frac{100 \times 1.35^2}{47.7^2 \times 0.025} = 3.20m$

$3.20 \simeq 3.22 \Rightarrow \Delta h = 3.2m$ kabul edilir

$3.22 \neq 3.40 \Rightarrow \Delta h_{2.it} = 3.22$ (3. iterasyon)

$k = 3 \times 10^{-4} < \delta_a = 1.3 \times 10^{-4}$: Hydraulically Smooth

c) $\nu = 10^{-4}\ m^2/s$: $Re = VD/\nu = \frac{1.35 \times 0.1}{10^{-4}} = 1350 < 2320$: Laminar

Laminar flow → Hagen Poiseuille (Chezy türbülant için)

$V = \frac{gI}{\nu} \frac{D^2}{32}$, $\Delta h_L = \frac{64}{Re} \frac{L}{D} \frac{V^2}{2g} = 4.32m$ ← Darcy-Weisbach Eq. (λ)

ν (m²/s)	Re	Regime	Δh over 100 m
10^{-7}	1.35×10^4	Turbulent, hydraulically rough	2.5 m
10^{-5}	1.35×10^4	Turbulent, hydraulically smooth	3.2 m
10^{-4}	1350	Laminar	4.32 m

Problem 7.2: In a welded steel oil line (diameter 0.6m), pumps are spaced 10 km apart. The pumps are working a difference in pressure head of 25 m oil column. Calculate the discharge. Density of the oil is $\rho=800\ kg/m^3$, kinematic viscosity is $\nu=10^{-4}\ m^2/s$, boundary roughness is $a=5\times10^{-5}$ m. In the course of time, the roughness of the pipe (expressed in terms of roughness length will grow 3×10^2 times the original value. Will the discharge be influenced by this? If so, how much (in %) will the reduction (approximate) be?

$D = 0.6m$; $L = 10^4 m$; $\Delta h = 25m$; $\rho = 800 kg/m^3$; $\nu = 10^{-4} m^2/s$; $a = k/2 = 5 \times 10^{-5} m$; $a' = 300a$

1. $J = \frac{\Delta h}{L} = \frac{25}{10^4} = 2.5 \times 10^{-3}$, $R = D/4 = 0.15m$

! bu noktada, turbulent flow kabulü yaparak soruyu çözeceğiz, en son bu kabulün doğrulağunu kontrol edeceğiz.

2. $u_* = \sqrt{gRJ} = \sqrt{10 \times 0.15 \times 2.5 \times 10^{-3}} = 0.061 m/s$

   $\delta_a = 11.6\,\nu / u_* = 11.6 \frac{10^{-4}}{0.061} \simeq 0.02m$

   $k = 10^{-4} m << \delta_a (= 0.02m)$ ⇒ hydraulically smooth (ν'ye bağlı (a veya k yok))

3. $C = 18 \log \frac{6R}{\delta_a/7} = 18 \log \frac{6 \times 0.15}{0.02/7} = 65\ m^{1/2}/s$

4. $V = C\sqrt{RJ} = 65\sqrt{0.15 \times 2.5 \times 10^{-3}} = 0.87 m/s$

   $Q_1 = V \frac{\pi D^2}{4} \simeq 0.246 m^3/s$

   $Re = VD/\nu = \frac{0.87 \times 0.6}{10^{-4}} = 5200 > 2320$: Turbulent ✓

~ old pipe : $a' = 300a = 15 \times 10^{-3} m$; $k' = 2a' = 3 \times 10^{-2} m$

Δh, J ve δ_a aynı , sinali transition'da ⇒ $\delta_a = 0.02 < k' = 0.03 < 6\delta_a = 0.12m$

1. Transition : C hesabında, a ve δ_a kullanılır.

   $C = 18 \log \frac{6R}{a' + \delta_a/7} = 18 \log \frac{6 \times 0.15}{15 \times 10^{-3} + 0.02/7} = 30.6\ m^{1/2}/s$

2. $V' = C\sqrt{RJ} = 30.6\sqrt{0.15 \times 2.5 \times 10^{-3}} = 0.53 m/s$

   $Q_2 = V' \frac{\pi D^2}{4} = 0.149 m^3/s$

$\frac{0.246 - 0.149}{0.246} \times 100 \simeq 32\%$ reduction

Problem 7.3: Find the discharge of the pipe system shown in Figure 7.3 and if the diameter of the system is converted into ϕ150 mm find the length of the pipe of the system.

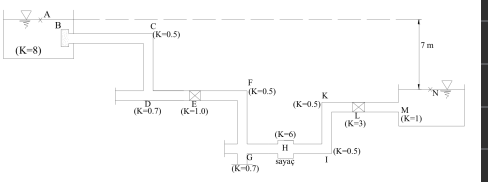


Figure 7.3

Between B-G
$L = 40m$
$\lambda = 0.025$
$D = 300mm$

Between G-M
$L = 30m$
$\lambda = 0.030$
$D = 150mm$

1. Energy eq. between two points: $\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 + H_p = \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 + H_T + \Sigma \Delta h$

   → $\frac{p_M}{\gamma}^{0} + \frac{V_M^2}{2g}^{0} + z_M = \frac{p_G}{\gamma}^{0} + \frac{V_G^2}{2g}^{0} + z_G + \Sigma \Delta h$ ⇒ $\Sigma \Delta h = z_M - z_G = 3m$

   friction: $\lambda \frac{L}{D} \frac{V^2}{2g}$ (Darcy-Weisbach) — toplu boyunca sürtünmeden kaynaklanan enerji kaybı (major loss)

   local loss: $K \frac{V^2}{2g}$ — toplam enerji kaybı : $\Sigma \Delta h$ = major loss + minor loss

   sudden expansion: $\frac{(V_1 - V_2)^2}{2g}$

2. 7 'şer kor bir boruya bölü: $\Sigma K_1 = 10.7$, $\Sigma K_2 = 11.7$

   $\Sigma \Delta h = \frac{V_1^2}{2g}\left(\Sigma K_1 + \lambda_1 \frac{L_1}{D_1}\right) + \frac{V_2^2}{2g}\left(\Sigma K_2 + \lambda_2 \frac{L_2}{D_2}\right)$

3. $\Phi = V_1 A_1 = V_2 A_2 \Rightarrow V_1 \times \frac{\pi \times 0.3^2}{4} = V_2 \times \frac{\pi \times 0.15^2}{4} \Rightarrow V_1^2 = V_2^2 / 16$,

   $7 = \frac{V_2^2}{2g}\left[\frac{1}{16}\left(8 + 0.5 + 0.7 + 10.5 + 0.025 \frac{46}{0.3}\right) + \left(0.7 + 1 + 0.5 + 2 + 1 + 0.02 \frac{30}{0.15}\right)\right] = 16.66 \frac{V_2^2}{2g}$ ⇒ $V_2 = 2.3 m/s$ (velocity in the ϕ150mm pipe)

   ($\lambda_1 L_1 / D_1$, $\lambda_2 L_2 / D_2$)

   $\Phi = 2.3 \times \frac{\pi \times 0.15^2}{4} = 0.05 m^3/s$

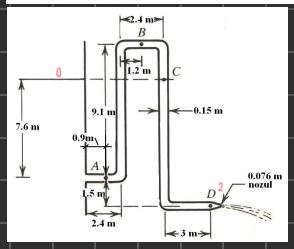
4. Tüm boruya ϕ150mm boruyla değiştirirsek, aynı Φ ve Δh değerleriyle hesaplarsak uzunluğu kaç olmalı?

   $7 = \lambda_2 \frac{L_e}{D_2} \frac{V_2^2}{2g} = 0.020 \times \frac{L_e}{0.15} \times \frac{2.88^2}{19.62} \Rightarrow L_e = 13.5m$

* Burada head loss'u yalnızca Darcy-Weisbach formülüne eşitliyoruz çünkü enerji kaybı yalnızca sürtünmelerden kaynaklı olacak, minor loss'lar dirsek, vana, sayaç gibi elemanlardan kaynaklanır.
  └ biz burada sistemi basitleştirerek hepsini boruya indirgedik.

Problem 7.4: Determine the discharge for system as given in Figure and calculate the pressures at Points A, B, C, D. Head loss per unit length and head loss at the nozzle may be calculated with using equations of $\left(\frac{0.1\,V^2}{2g}\right)$ and $\left(\frac{3V^2}{2g}\right)$, respectively. Neglect the head losses at elbows and entrance.

D = 0.15m , d = 0.076m
pipe length : 27.5m



1. 0 → 2 : enerji denklemi, $\frac{P_0}{\gamma}+\frac{V_0^2}{2g}+z_0=\frac{P_2}{\gamma}+\frac{V_2^2}{2g}+z_2+\Sigma h_L$

$z_0-z_2=\frac{V_2^2}{2g}+\Sigma h_L \Rightarrow 7.6+1.5=\frac{V_2^2}{2g}+fL+\frac{2V_2^2}{2g}$

$9.1=\frac{4V_2^2}{2g}+0.1\times27.5\frac{V_2^2}{2g}$ → $V_2=6.53$m/s, $V_1=1.67$m/s, $Q=0.0237$m³/s

$V_2(0.15)^2=V_1(0.076)^2 \Rightarrow V_1=0.25V_2$

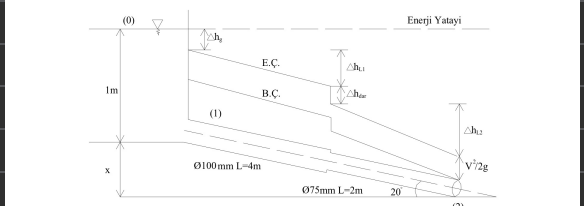
2. (0-A) $\frac{P_0}{\gamma}+\frac{V_0^2}{2g}+z_0=\frac{P_A}{\gamma}+\frac{V_A^2}{2g}+z_A+\Sigma h_L \Rightarrow 7.6=\frac{P_A}{\gamma}+\frac{1.67^2}{2g}+0.9\times0.1\times\frac{1.67^2}{2g} \Rightarrow P_A=34.43$ kPa

(0-B) $\frac{P_0}{\gamma}+\frac{V_0^2}{2g}+z_0=\frac{P_B}{\gamma}+\frac{V_B^2}{2g}+z_B+\Sigma h_L \Rightarrow 0=\frac{P_B}{\gamma}+\frac{1.67^2}{2\times9.81}+1.5+\left(0.1\frac{1.67^2}{2g}\right)\times(2.4+3.1+2\times1.5)$ ($V_{pipe}=1.67$m/s) $\Rightarrow \frac{P_B}{\gamma}=-0.143-1.5-0.182-1.83m \Rightarrow P_B=-13.5$kPa

(0-C) $\frac{P_0}{\gamma}+\frac{V_0^2}{2g}+z_0=\frac{P_C}{\gamma}+\frac{V_C^2}{2g}+z_C+\Sigma h_L \Rightarrow 0=\frac{P_C}{\gamma}+\left(2.4+3.1+2\times1.5+\left(0.1\times\frac{1.67^2}{2g}\right)\right)\times\frac{1.67^2}{2g} \Rightarrow P_C=-3.44$kPa

(0-D) $\frac{P_0}{\gamma}+\frac{V_0^2}{2g}+z_0=\frac{P_D}{\gamma}+\frac{V_D^2}{2g}+z_D+\Sigma h_L \Rightarrow 7.6+1.5=\frac{P_D}{\gamma}+\frac{1.67^2}{2g}+23.5\times0.1\times\frac{1.67^2}{2g} \Rightarrow P_D=34.4$kPa

Problem 7.5: A fluid with a kinematic viscosity $\nu=15\times10^{-7}$ m²/s flows in the pipe system in Figure 7.5. The roughness of the pipe is k=0.001 m and the minor head loss coefficients of the system are known as $K_{contraction}$=0.21 $K_{entrance}$=0.05. Find the discharge of the system by using Moody diagram.



1. $\frac{P_0}{\gamma}+\frac{V_0^2}{2g}+z_0=\frac{P_2}{\gamma}+\frac{V_2^2}{2g}+z_2+\Sigma h_L \Rightarrow 1+x=\frac{V_2^2}{2g}+\Sigma h_L$ $\quad x=(4+2)\sin30=2.85$m

$3.05=\frac{V_2^2}{2g}+K_c\frac{V_1^2}{2g}+\lambda_1\frac{L_1V_1^2}{D_1 2g}+K_c\frac{V_2^2}{2g}+\lambda_2\frac{L_2V_2^2}{D_2 2g}$

* Sürtünme katsayılarını (λ) ve hızları bilmiyoruz.

$\lambda=f(Re,\ \varepsilon/D)$; $Re=VD/\nu$ → bu nedenle moody diyagramını kullanarak bir kabulde bulunuyoruz : fully rough

Re bilmek için de V'yi bilmemiz gerektiğinden fully rough kabulü yaptık, sürtünü boruda λ değeri sadece ε/D ye bağlıdır. (Çizgiler düz)

$\varepsilon/D_1=1/100=0.010 \rightarrow \lambda_1=0.038$

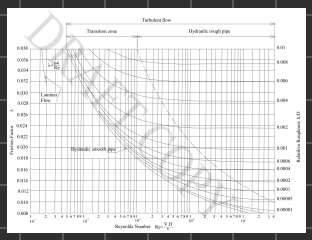
$\varepsilon/D_2=1/75=0.013 \rightarrow \lambda_2=0.042$

$\Theta_1=\Theta_2 \Rightarrow V_1A_1=V_2A_2 \Rightarrow V_2=1.78V_1$, $\frac{V_2^2}{2g}=\frac{1.78^2V_1^2}{2g}=3.16\frac{V_1^2}{2g}$

$3.05=\frac{V_2^2}{2g}+K_c\frac{V_1^2}{2g}+\lambda_1\frac{L_1V_1^2}{D_1 2g}+K_c\frac{V_2^2}{2g}+\lambda_2\frac{L_2V_2^2}{D_2 2g}$

$3.05=\frac{V_1^2}{2g}\left(3.16+0.05+0.038\times\frac{4}{0.1}+0.21\times3.16+0.042\times\frac{2}{0.075}\times3.16\right)$

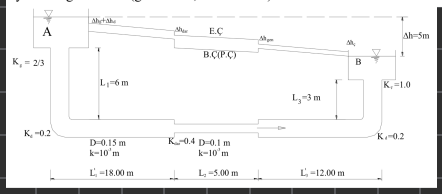
$3.05=8.32\frac{V_1^2}{2g} \Rightarrow V_1=2.53$m/s ; $V_2=4.5$m/s $\Rightarrow Q=0.012$m³/s



* kabulün kontrolü (h. rough) : $Re_1=VD/\nu=\frac{2.53\times0.1}{1.5\times10^{-6}}=1.73\times10^5$

$Re_2=VD/\nu=\frac{4.5\times0.075}{1.5\times10^{-6}}=2.30\times10^5$ → moody üzerinden kontrol ✓

Problem 7.6: Water flows from the reservoir A to reservoir B in the Figure 7.6. The elevation difference between the reservoirs is Δh=5.00 m. Find the discharge. Draw the energy and hydraulic grade line. (g=10 m/s², ν=10⁻⁶ m²/s)



1. $\frac{P_A}{\gamma}+\frac{V_A^2}{2g}+z_A=\frac{P_B}{\gamma}+\frac{V_B^2}{2g}+z_B+\Sigma h_L$

$5=\Sigma h_L=K_g\frac{V_1^2}{2g}+\lambda_1\frac{L_1}{D_1}\frac{V_1^2}{2g}+K_g\frac{V_1^2}{2g}+K_{ani}\frac{V_2^2}{2g}+\lambda_2\frac{L_2}{D_2}\frac{V_2^2}{2g}+\frac{(V_2-V_3)^2}{2g}+\lambda_3\frac{L_3}{D_3}\frac{V_3^2}{2g}+K_g\frac{V_3^2}{2g}+K_k\frac{V_3^2}{2g}$

minor loss (local loss) — major loss (friction) (Darcy-Weisbach) — sudden contraction — sudden expansion (diğer sorularda ani genişleme gibi bunun da kaybının emiliyoruz $K_{exp}\frac{V_2^2}{2g}$ şeklinde kullanılır)

2. Chezy eq. : $V=C\sqrt{RI} \rightarrow I=\frac{V^2}{C^2R}$

Darcy-Weisbach eq. : $I=\frac{\Delta h_L}{L}=\lambda\frac{V^2}{2gD}$ } $\lambda=\frac{2gD}{C^2R}$ ($R=D/4$) ⇒ turbulent flowların kapsında geçerli

$C=18\log\left(\frac{6R}{\varepsilon+3/3}\right) \Rightarrow C=18\log\frac{6R}{\varepsilon}$ * hydraulically rough assumption

$\rightarrow C_1=18\log\frac{6\times(0.15/4)}{10^{-3}/2}=47.36\sqrt{m}/s$, $\lambda_1=\frac{8\times9.81}{47.36^2}=0.035$; $C_3=C_1=47.36\sqrt{m}/s$, $\lambda_3=\lambda_1=0.035$

$\rightarrow C_2=18\log\frac{6\times(0.1/4)}{10^{-3}/2}=44.53\sqrt{m}/s$, $\lambda_2=\frac{8\times9.81}{44.53^2}=0.04$ $\quad \Theta_1=\Theta_2=\Theta_3 \Rightarrow V_1A_1=V_2A_2 \Rightarrow V_2=2.25V_1$

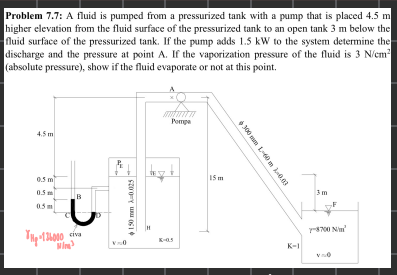
3. $5=\frac{2}{\gamma}\frac{V_1^2}{2g}+0.035\times\frac{24}{0.15}\times\frac{V_1^2}{2g}+0.2\times\frac{V_1^2}{2g}+0.4\times\frac{5V_1^2}{2g}+0.84\times\frac{5}{0.1}\times\frac{5V_1^2}{2g}+1.25^2\times\frac{4V_1^2}{2g}+0.035\times\frac{15}{0.15}\times\frac{V_1^2}{2g}+0.2\times\frac{V_1^2}{2g}+\frac{V_1^2}{2g}$

$V_1=2.02$m/s $\quad Q=2.02\times\frac{\pi\times0.15^2}{4}=0.0357$m³/s $\quad V_2=4.5$m/s

Control : $\gamma_1=\lambda_1\frac{V_1^2}{D_1 2g}=0.035\times\frac{2.02^2}{0.15\times19.62}=0.048$

$V_{c_1}=\sqrt{gD}=\sqrt{10\times0.15}\times0.048=0.135$m/s

$\delta_1=11.6\nu/V_{c_1}=\frac{11.6\times10^{-6}}{0.135}=8.6\times10^{-5}$



Problem 7.7: A fluid is pumped from a pressurized tank with a pump that is placed 4.5 m higher elevation from the fluid surface of the pressurized tank to an open tank 3 m below the fluid surface of the pressurized tank. If the pump adds 1.5 kW to the system determine the discharge and the pressure at point A. If the vaporization pressure of the fluid is 3 N/cm² (absolute pressure), show if the fluid evaporate or not at this point.

1. $P_C=P_0$;

$P_C=P_0+\gamma_{fluid}(0.5)$

$P_0=P_E+\gamma(1.5)$

$P_E=\gamma_{fluid}(0.5)-\gamma(1.5)=54\,950$ N/m²

2. energy eq. E-F (pump head)

$\frac{P_E}{\gamma}+\frac{V_E^2}{2g}+z_E+H_{PM}=\frac{P_F}{\gamma}+\frac{V_F^2}{2g}+z_F+\Sigma h_L$

$\frac{54\,950}{8700}+3+H_{PM}=\Sigma h_L \Rightarrow 9.32+H_{PM}=\Sigma h_L$

3. $\Sigma h_L=\Sigma h_{minor}+\Sigma h_{major}=0.5\frac{V_{AH}^2}{2g}+0.025\frac{15}{0.15}\frac{V_{AH}^2}{2g}+0.03\frac{60}{0.1}\frac{V_{AF}^2}{2g}+1\frac{V_{AF}^2}{2g}=2\frac{V_{AH}^2}{2g}+3\frac{V_{AF}^2}{2g}=2.8V_{AF}^2$

(Darcy-Weisbach eq.)

$\rightarrow Q=V_{AH}A_{AH}=V_{AF}A_{AF} \Rightarrow V_{AH}=4V_{AF}$

4. the pump power : $N=\frac{\gamma Q H_{PM}}{\eta}=1.5$ ($\gamma=8700\,\frac{N}{m^3}$) $\Rightarrow H_{PM}=\frac{0.172}{\Theta_1}$; $9.32+\frac{0.172}{\Theta_1}=2.8\left(\frac{\Theta_1}{\pi\times0.1^2/4}\right)^2 \Rightarrow 560.4\Theta_1^3-9.32\Theta_1-0.172=0$

$\Theta_1=0.139$ m³/s

(1.5 kW pompanın sağladığı enerji olarak belirtildiği için iletimde olmadığı bir durum yok)

5. A noktasındaki basınç bulmak için E-A arası enerji denklemini tekrar yazalım ; $z_E+\frac{P_E}{\gamma}+\frac{V_E^2}{2g}+z_A+\frac{P_A}{\gamma}+\frac{V_A^2}{2g}=\Sigma h_L$ ($\frac{54\,950}{8700}=6.31$m ; $\frac{0.5V_{AH}^2}{2g}+\frac{0.025}{0.15}\frac{V_{AH}^2}{2g}$)

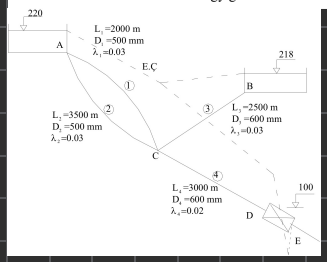
$V_{AH}=\frac{\Theta_1}{A_{AH}}=\frac{0.139}{\pi\times0.15^2/4}=7.35$m/s

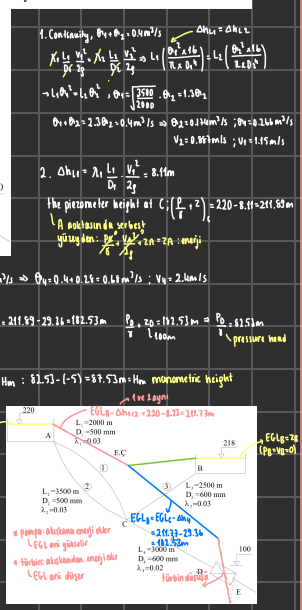
$\rightarrow 6.31+4.5+\frac{P_A}{\gamma}+\frac{V_A^2}{2g}+3\frac{V_{AH}^2}{2g} \Rightarrow \frac{P_A}{\gamma}=-10.42$m , $P_A=-10.42\times8700=-90\,692.93$ N/m² (relative Pressure)

$(P_A)_{absolute}=P_{atm}+(P_A)_{gage}=100\,000+(-90\,692.93)=9\,307.13$ N/m²

$(P_A)_{absolute}=9\,307.13$ N/m² < 3 N/cm² : the fluid evaporates

Problem 7.8: The discharge from the reservoir A to reservoir C is 400 lt/s in Figure 7.8. The pressure at the outlet of the turbine DE is –5 mwc. Find the power which could be obtained from the turbine and draw the energy grade line of the system.



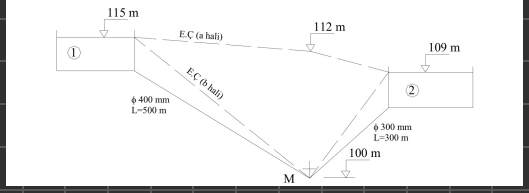


Problem 7.9: The fluid flows from reservoirs (1) and (2) to valve at M. The Manning roughness coefficient for the pipes are n=0.014.

a) The piezometer height at M is 112 m,

b) And when the valve is opened at M determine the discharge.

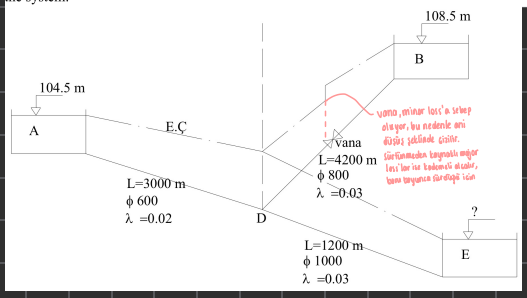
c) Draw the energy grade line for both cases.



a) H1 = 115m, Hm = 112m, H2 = 109m ⇒ H1 > Hm > H2 akış yönü ⇒ Θm = Θ1 − Θ2
Manning eq.: Θ = (1/n) R^(2/3) J^(1/2) A
Θ1 = (1/0.014)((0.4/4)^(2/3))((115−112)/500)^(1/2)((π(0.4)²)/4) = 0.49 m³/s
Θ2 = (1/0.014)((0.3/4)^(2/3))((112−109)/300)^(1/2)((π(0.3)²)/4) = 0.89 m³/s
ΘM = Θ1 − Θ2 = 0.40 m³/s = 401 l/s

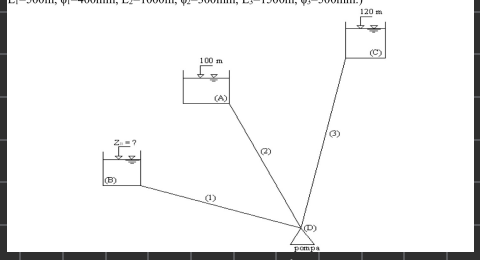
b) M vanası açıldığında, pM/γ = 0 ; HM = pM/γ + zM = 100m
H1 = 115m, H2 = 109m, HM = 100m ⇒ H1 > H2 > HM ⇒ akış 1→M, 2→M
Θ1 = (1/0.014)((0.4/4)^(2/3))((115−100)/500)^(1/2)((π(0.4)²)/4) = 0.335 m³/s
Θ2 = (1/0.014)((0.3/4)^(2/3))((109−100)/300)^(1/2)((π(0.3)²)/4) = 0.156 m³/s
Θ1 = Θ1 + Θ2 = 0.491 m³/s

Problem 7.10: Determine the discharge at each pipe when the friction head loss in the pipe AD is 14 m in Fig 7.10 and find the elevation of the reservoir E, draw the energy grade line of the system.

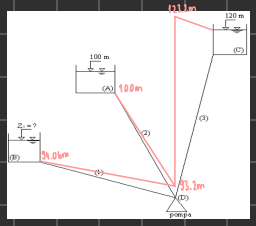


1. ΔhAD = 14 = λAD (lAD/DAD)(VAD²/2g) ⇒ VAD = 1.66 m/s , ΘAD = VAD AAD = 1.66 x π x 0.8²/4 = 0.83 m³/s , (p/γ + z)D = 104.5 − 14 = 90.5m
HA = 104.5m, HB = 108.5m, HD = 90.5m ⇒ HB > HD > HA ⇒ akış
2. ΔhBD = 108.5 − 90.5 = 18m = λBD (lBD/DBD)(VBD²/2g) ⇒ VBD = 1.5 m/s ⇒ ΘBD = VBD ABD = 0.754 m³/s
3. ΘDE = ΘBD + ΘAD = 0.83 + 0.754 = 1.222 m³/s = VDE π x 1.2²/4 ⇒ VDE = 1.553 m/s
4. ΔhLDE = λDE (lDE/DDE)(VDE²/2g) = 0.03 x (1200/1) x (1.553²/19.62) = 4.45m
zE = (p/γ + z)D − ΔhLDE = 90.5 − 4.45 = 86.05m
Ekin, V=0 ; P=0

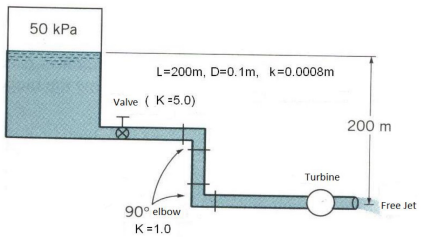
Problem 7.11: Reservoir A and B are feeding reservoir C by a pump located at point D. The discharges from Reservoir A and B to Reservoir C have same discharges as 100 lt/sec. If the efficiency of the pump is η = 0.70, what will be required power of this pump. Determine the elevation of Reservoir B. Draw the energy grade line of the system. (λ=0.02, η=0.70, L1=500m, ϕ1=400mm, L2=1000m, ϕ2=300mm, L3=1500m, ϕ3=500mm.)



1. ΘE = Θ2 = 0.1 m³/s ⇒ 0.1 = V1 x π x 0.4²/4 ⇒ V1 = 0.8 m/s ; 0.1 = V3 x π x 0.5²/4 ⇒ V3 = 1.4 m/s
Θ1 + Θ2 = Θ3 = 0.2 m³/s ⇒ 0.2 = V2 x π x 0.3²/4 ⇒ V2 = 1.8 m/s
2. A-C: pA/γ + vA²/2g + zA + Hm = pC/γ + vC²/2g + zC + (L1/D1)(V1²/2g) + (L3/D3)(V3²/2g) ⇒ zA + Hm = zC + ΔhL ⇒ Hm = 29.94m
(pressure by the pump, major losses (due to friction))
3. B-C: pB/γ + vB²/2g + zB + Hm = pC/γ + vC²/2g + zC + ΔhL ⇒ 29.94 + zB = 120 + 0.2 (500/0.4)(0.8²/19.62) + (1500/0.5)(1.8²/19.62) ⇒ zB = 94.06m
4. N = γ Θ4 Hm / η = 10 x 0.2 x 29.94 / 0.7 = 85.54 kW



Problem 7.13: Water flows through the pipe system as seen in Figure below. If the manometric height of the turbine is 116 m, determine the discharge in the pipe system. (Neglect head loss at the exit of the tank). ($\nu=10^{-6}$ m^2/s)



$\frac{P_1}{\gamma}+\frac{V_1^2}{2g}+z_1=\frac{P_2}{\gamma}+\frac{V_2^2}{2g}+z_2+H_m+\Sigma h_L \rightarrow \frac{50}{100}+200=\frac{V_2^2}{2g}+116+\frac{V^2}{2g}\left(\lambda\frac{200}{0.1}+5+1\right) \Rightarrow 89=(8+2000\lambda)\frac{V^2}{2g}$

$\frac{P_1}{\gamma}+(z_1-z_2)=\frac{V_2^2}{2g}+H_m+\Sigma h_L$; Σh_L → major: $\lambda\frac{L}{D}\frac{V^2}{2g}$; minor: $K\frac{V^2}{2g}$

$\Rightarrow Q=AV=\pi\frac{0.1^2}{4}\times 4.68=0.0367\ m^3/s$

Amacımız debiyi bulmak (Q), bunun için V'yi bulmamız gerekiyor, onra V'yi bulmak için de λ'yı bilmemiz gerekiyor. Bunu da Moody üzerinden bulucaz.

Relative Roughness: $\varepsilon/D=0.0008/0.1=0.008$ → (Çok yüksek → hyd. rough tabiiki yapıyoruz)

$\lambda=0.0358$ (yaptığımız kabule uygun değer)

$89=(8+0.0358\times 2000)\frac{V^2}{2g}$ ⇒ $V_2=4.68\ m/s$ → R_e hesabı yaparak kabullerimizi doğru mu değil mi tespit edeyim.

$Re=4.68\times 0.1/10^{-6}=468\,000$ ✓ Hydraulically Rough